Aanchal Singh

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ABOUT

M.Tech Biotech student at NIT Jalandhar, with strong hands-on experience in metagenomics, biodegradable biomaterials, and natural product chemistry. Skilled in molecular biology techniques, bioinformatics tools, and in-silico analysis, with a solid foundation in both wet-lab and computational research. Demonstrates strong research aptitude, problem-solving, and teamwork abilities, with a focused interest in sustainable, environmental, and biomedical biotechnology solutions.

EDUCATION

Dr B. R Ambedkar National Institute of Technology, Jalandhar — *Jalandhar, Punjab*
M.Tech (Biotechnology) CGPA: 7.93 — *August, 2025 - Present*

Bundelkhand University — *Jhansi, U.P.*
B.Tech (Biotechnology) CGPA: 8.4 (Ranked 2nd in B.Tech. Biotechnology) — *October 2021 - June 2025*

WORK EXPERIENCE

Research Intern: Ministry of Environment, Forest and Climate Change (MoEFCC) — *Dehradun, Uttarakhand*
June 2024 - Aug 2024

- Conducted environmental monitoring and analysis aligned with biodiversity conservation.
- Gained exposure to field and lab techniques in ecological and environmental studies.

Research Trainee, Department of Molecular and Human Genetics, BHU — *Varanasi, U.P.*
June 2023 – July 2023

- Received hands-on training in immunogenetics and molecular biology techniques.
- Performed DNA/RNA isolation, electrophoresis, and gene expression studies under expert supervision.

PROJECTS

Optimization and Development of Potato Peel Biofilm: A Waste Valorisation Approach Against Plastic Pollution

- Designed eco-friendly plastic alternatives using agricultural biomass.
- Enhanced film strength and biodegradability through biopolymer modification.

Metagenomic Analysis of Microbial and Functional Dynamics in the Godavari River: Impacts of Anthropogenic Activities - *Supervised by Dr Vipin Gupta (Scientist C, MoEFCC, Dehradun, Uttarakhand)*

- Studied microbial diversity during large-scale religious events.
- Used metagenomics tools for taxonomy classification and functional annotation.
- Assessed the ecological impacts of anthropogenic activities on water bodies.

Comparative analysis of NMIIA mutations to evaluate their effects on flavonoid binding and S100A4 interaction using molecular docking and structural analysis, highlighting mutation- driven changes linked to cancer progression - ***Major Project***

- To compare wild-type and mutant NMIIA proteins assessing their binding to anticancer flavonoids and their interaction with S100A4 via molecular docking and structural analysis in order to reveal mutation-driven changes relevant to cancer progression.

SKILLS

- **Laboratory Techniques:** DNA isolation, RNA extraction, gel electrophoresis, spectrophotometry, microbial culturing, biochemical estimations, sterilization techniques.
- **Bioinformatics Tools:** QIIME, MEGA, NCBI BLAST, Linux command-line tools.
- **Software Proficiency:** Microsoft Office Suite (Word, Excel, PowerPoint), Canva.
- **Programming Scripting:** Basic command-line scripting in Linux for data processing.
- **Soft Skills:** Scientific communication, problem-solving, critical thinking, teamwork, leadership.

PUBLICATIONS

Optimization and characterization of potato peel bioplastic: A green chemistry approach against plastic pollution - *Journal of Hazardous Materials: Plastics, 4 (2026), 100076* *Link*

- Co-authored a peer-reviewed research publication on the development and optimization of sustainable bioplastic from potato-peel waste using Response Surface Methodology (RSM).
- Contributed to experimental analysis and data curation, including evaluation of biodegradability, surface morphology, chemical characteristics, and mechanical properties.
- The optimized formulation demonstrated **46–52% biodegradation within 2 weeks** and **6.07 MPa tensile strength**, highlighting its potential as an eco-friendly alternative to conventional plastic.

EXTRACURRICULAR ACTIVITY

Certifications & Courses

- GATE Biotechnology 2024 - Qualified
- NPTEL: Bioreactor Design & Analysis, Computational Genomics
- Hands-on Workshop: Intellectual Property Rights & Genomic Assembly (AMI, NABARD, ICAR)

Conferences & Presentations

- International Symposium, IIT Roorkee - Best Oral Presentation Award
- National Conference on Plant-Microbe Interactions - Poster Presentation
- National Seminar, Maha Kumbh 2025 - Oral Presentation

REFERENCES

Dr. Mahesh Kumar Sah

Assistant Professor, NIT Jalandhar, sahmk@nitj.ac.in

Dr. Anand Kumar Pandey

Assistant Professor, Bundelkhand University, dr.akpandey@bujhansi.ac.in